A-A-50759
15 MAY 86

COMMERCIAL ITEM DESCRIPTION

SIX-OUTPUT SWITCHING POWER SUPPLY

The General Services Administration has authorized the use of this Commercial Item Description.

Scope. This Commercial Item Description (CID) specified a six-output, 451 watt power supply which features regulated ± 30 VDC and ± 65 VDC output for circuits interface assembly in the Fire Control Radar Track and System Control Test Station. This system shall be known as Six-Output Switching Power Supply.

Applicable documents.

MIL-M-7298 - Manual, Technical, Commercial Equipment.
MIL-T-28800 - Test Equipment for use with Electrical and Electronic Equipment, General Specification for.
MIL-M-38797 - Manual Technical Operation Instructions and Maintenance Instructions (for various Typing Equipment).

Salient characteristics.

1. Performance requirements.

a. Six-output switching power supply. The power supply has six output modules consisting of minimum of 451 watts switching. Output voltage is protected against overload and short circuit by an electronic current limiting fold back circuit with automatic reset upon removal of the short circuit or overload. The power supply also has an internal fan with a thermal sensor activation in the event an overheat condition occurs due to a failure. The power supply shall be capable of providing a constant voltage output which can be set at an input of 115-230 volts AC with 47-63 Hz single phase frequency range. The current ratings available for each nominal voltage are as follows:

Output #1	Output #2	Output #3	Output #4	Output #5	Output #6
30VDC@1A	33VDC@1A	65VDC@2.5A	-30VDC@1A	-33VDC@1A	-65VDC@2.5A

The power supply shall have the complete facility for calibrating at the above ranges of power.

b. Voltage accuracy. The input line regulation shall have an accuracy of 0.1 percent for ± 15% line change. The regulation load shall have ± 2% accuracy from a minimum load to a full load change. Voltage change is 0.02% per ^{o}C.

c. Voltage stability. The ripple and noise of the output voltage shall not vary more than 1 percent P-P to 20 MHz periodic and random deviation.

FSC 6130

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d. Temperature coefficient. The power supply temperature coefficient should be 0.03 percent per degree C.

e. Operating temperature. The operating temperature for the power supply is 0 to 50°C full rated. It is derated at 50% to 70°C.

f. Storage temperature. The storage temperature of the power supply is -55 to 85°C.

g. Operating altitude. The operating altitude is 0 to 10,000 feet full rated.

h. Collateral power output indicator. A collateral power output indicator shall be provided for monitoring the output voltage and current. This output power indicator shall be required to provide a relative, convenient, monitor to directly track the output power within an accuracy of ± 10 percent.

2. Type and size.

a. Design and construction. Unless otherwise specified herein, the equipment shall comply with the type III design and construction requirements specified in MIL-T-28800.

b. Solid state construction. The equipment shall be of solid-state modular miniaturized construction. Electron tubes shall not be used, except that high power RF signal production may necessitate the use of electronic vacuum tubes in the output states.

c. Receptacle and connector mounting. All AC input and DC output connectors are provided on barrier strip (#6 screws) for jumper wire assembly. The connectors shall comply with MIL-T-28800 3.4.1.5.2.

d. Voltage adjustment. All output voltage shall be externally adjustable ± 10 percent. The ± 30 VDC and the ± 33 VDC shall have a single tracking adjustment for each, but the ± 65 VDC must have individual voltage adjustments.

e. Accessibility. The equipment shall be constructed so subassemblies and chassis components can be removed without removing any other hardware subassembly, printed circuit card or component.

f. Input power switch. The input power switch shall comply with type III input switch requirements of MIL-T-28800, except that the switch shall break both sides of the power source, but not break the earth ground conductor circuit.

g. Fuses or circuit breakers. Fuses or circuit breakers shall comply with type III requirements of MIL-T-28800, except that both sides of the input power line shall be equipped with fuses or circuit breakers.

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3. Dimensions, weight and mechanical stability.

a. Weight. The unpacked weight of the power supply, including all accessories, shall not exceed 14 pounds.

b. Dimensions. The maximum dimensions of the power supply, including all accessories, installed in the style E enclosure as referenced in MIL-T-28800 shall not exceed 7 1/2 inches in width, 5 inches in height and 11 1/8 inches in depth. The dimensions shall be identified in Metric and English form.

Certification. The contractor shall certify that the product offered meets the salient characteristics of this description, and that the product conforms to the producer's own drawings, specifications, standards and quality assurance practices. Bid samples from bidders will be tested in accordance with the salient characteristics of this Commercial Item Description at San Antonio ALC, Kelly AFB, TX 78241.

Regulatory requirements. The offeror/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent practical.

Packaging, packing and marking. The items shall be packaged in accordance with normal commercial practice and packed to assure acceptance by common carrier and provide product protection against loss and damage during multiple shipments, handling, and storage. The shipping container shall be in compliance with national motor freight classification and uniform freight classification. Shipping containers shall be marked as specified in the contract or order.

Intended use. The equipment shall be designed for use at Air Force Base Depot Precision Measuring Equipment Laboratories. This item must be equal or better to the P/N SM3-5079, manufacturer Todd Products Corp.

Known acceptable products.

Manufacturer	Model	NSN
Todd Products Corp No longer available	P/N SM3-5079	6130-01-122-8712DQ

Operational instructions/maintenance manual. The contractor shall prepare an operator's manual according to the provisions of MIL-M-38797 for all equipment delivered under terms of this contract.

Commercial publications. Commercial publications may be submitted to fulfill Air Force documentation requirements. These publications require technical order numbering and must be referenced in the system level publication. Such commercial data must be prepared in accordance with MIL-M-7298C. Two copies of each commercial manual proposed shall be submitted for Air Force evaluation and approval. Commercial publications shall be reviewed by the Technical Order Management Office (TOMO) before final acceptance by the Air Force. If modifications are performed on Air Force equipment, the contractor shall provide

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updating manual changes to meet Air Force standards. All applicable reproduction rights for Government use must accompany submittals.

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Preparing activity:
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